

Topic 8: Grey matter

The scene is set by considering how the working of the nervous system enables us to see. Brain imaging and the regions of the brain are considered. The topic also demonstrates how an understanding of brain structure and functioning is relevant to such issues as the response to stimuli, the development of vision and learning. It investigates how imbalances in brain chemicals may result in conditions such as Parkinson's disease and its treatment with drugs are investigated. Students discuss the ethical issues raised by the Human Genome Project and the risks and benefits of using genetically modified organisms.

8.2 Assessment information

This unit is assessed by means of a written examination paper, which carries 90 marks, lasts 1 hour 45 minutes and will include:

- objective questions
- structured questions
- short-answer questions

and will also cover:

- *How Science Works*
- practical-related questions.

Students may use a calculator.

One question will relate to a previously released scientific article that students will have studied during the course. Students may be asked to summarise the information in the article, and explain or comment upon the biology and other issues within the context of the article. The article may draw on knowledge and understanding from any of the four units 1, 2, 4, and 5. A different article will be provided each year and the examination questions will change to reflect this. This question carries a third of the marks of this unit.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.